

Task 1: Inspect inodes, directories and links

Output:

```
karthik@Karthikeya: ~/ossp/v  + - x
karthik@Karthikeya:~$ mkdir -p ~/ossp/week10
karthik@Karthikeya:~$ cd ~/ossp/week10
karthik@Karthikeya:~/ossp/week10$ echo "File system practical" > file1.txt
karthik@Karthikeya:~/ossp/week10$ mkdir testdir
karthik@Karthikeya:~/ossp/week10$ ls -li file1.txt
13085 file1.txt
karthik@Karthikeya:~/ossp/week10$ stat file1.txt
  File: file1.txt
  Size: 22          Blocks: 8          IO Block: 4096   regular file
Device: 8,48   Inode: 13085         Links: 1
Access: (0644/-rw-r--r--)  Uid: ( 1000/  karthik)   Gid: ( 1000/  karthik)
Access: 2026-09-27 14:04:10.557302704 +0000
Modify: 2026-09-27 14:04:10.557302704 +0000
Change: 2026-09-27 14:04:10.557302704 +0000
 Birth: 2026-09-27 14:04:10.557302704 +0000
karthik@Karthikeya:~/ossp/week10$ find . -inum "$(ls -li file1.txt | awk '{print $1}')"
./file1.txt
karthik@Karthikeya:~/ossp/week10$ ln file1.txt hardlink.txt
karthik@Karthikeya:~/ossp/week10$ ln -s file1.txt symlink.txt
karthik@Karthikeya:~/ossp/week10$ ls -li
total 12
13085 -rw-r--r--  2 karthik karthik   22 Sep 27 14:04 file1.txt
13085 -rw-r--r--  2 karthik karthik   22 Sep 27 14:04 hardlink.txt
13186 lrwxrwxrwx  1 karthik karthik    9 Sep 27 14:08 symlink.txt -> file1.txt
13181 drwxr-xr-x  2 karthik karthik  4096 Sep 27 14:04 testdir
karthik@Karthikeya:~/ossp/week10$ stat file1.txt
  File: file1.txt
  Size: 22          Blocks: 8          IO Block: 4096   regular file
Device: 8,48   Inode: 13085         Links: 2
Access: (0644/-rw-r--r--)  Uid: ( 1000/  karthik)   Gid: ( 1000/  karthik)
Access: 2026-09-27 14:04:10.557302704 +0000
Modify: 2026-09-27 14:04:10.557302704 +0000
Change: 2026-09-27 14:08:15.034515628 +0000
 Birth: 2026-09-27 14:04:10.557302704 +0000
karthik@Karthikeya:~/ossp/week10$ stat hardlink.txt
  File: hardlink.txt
  Size: 22          Blocks: 8          IO Block: 4096   regular file
Device: 8,48   Inode: 13085         Links: 2
Access: (0644/-rw-r--r--)  Uid: ( 1000/  karthik)   Gid: ( 1000/  karthik)
Access: 2026-09-27 14:04:10.557302704 +0000
Modify: 2026-09-27 14:04:10.557302704 +0000
Change: 2026-09-27 14:08:15.034515628 +0000
 Birth: 2026-09-27 14:04:10.557302704 +0000
karthik@Karthikeya:~/ossp/week10$ stat symlink.txt
  File: symlink.txt -> file1.txt
  Size: 9           Blocks: 0          IO Block: 4096   symbolic link
Device: 8,48   Inode: 13186         Links: 1
Access: (0777/lrwxrwxrwx)  Uid: ( 1000/  karthik)   Gid: ( 1000/  karthik)
Access: 2026-09-27 14:08:27.707721441 +0000
Modify: 2026-09-27 14:08:20.122356434 +0000
Change: 2026-09-27 14:08:20.122356434 +0000
 Birth: 2026-09-27 14:08:20.122356434 +0000
karthik@Karthikeya:~/ossp/week10$ readlink symlink.txt
file1.txt
karthik@Karthikeya:~/ossp/week10$ cat hardlink.txt
File system practical
karthik@Karthikeya:~/ossp/week10$ cat symlink.txt
File system practical
```

Task 2: Read and write files using mmap()

Code:

```
karthik@Karthikeya: ~/ossp/v x + v
GNU nano 7.2 mmap_demo.c *
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>
#include <sys/mman.h>
#include <sys/stat.h>
#include <string.h>

int main() {
    const char *filename = "mmap_file.txt";

    int fd = open(filename, O_RDWR | O_CREAT | O_TRUNC, 0644);
    if (fd == -1) {
        perror("open");
        return 1;
    }

    const char *message = "Memory mapped file practical";
    size_t size = strlen(message);

    if (ftruncate(fd, size) == -1) {
        perror("ftruncate");
        close(fd);
        return 1;
    }

    char *ptr = mmap(NULL, size, PROT_READ | PROT_WRITE,
                     MAP_SHARED, fd, 0);

    if (ptr == MAP_FAILED) {
        perror("mmap");
        close(fd);
        return 1;
    }

    memcpy(ptr, message, size);

    if (msync(ptr, size, MS_SYNC) == -1)
        perror("msync");

    printf("Mapped file content: %.s\n", (int)size, ptr);

    munmap(ptr, size);
    close(fd);

    return 0;
}

?G Help      ?O Write Out  ?W Where Is   ?K Cut        ?J Execute    ?C Location   ?U Undo       ?A Set Mark   ?] To Bracket
?X Exit      ?R Read File  ?A Replace    ?U Paste      ?C Justify    ?/ Go To Line ?- Redo       ?-C Copy      ?_ Where Was
```

Output:

```
karthik@Karthikeya:~/ossp/week10$ nano mmap_demo.c
karthik@Karthikeya:~/ossp/week10$ gcc mmap_demo.c -o mmap_demo
karthik@Karthikeya:~/ossp/week10$ ./mmap_demo
Mapped file content: Memory mapped file practical
karthik@Karthikeya:~/ossp/week10$ cat mmap_file.txt
Memory mapped file practicalkarthik@Karthikeya:~/ossp/week10$ |
```